THE CO-OPERATIVE UNIVERSITY OF KENYA

SCHOOL: **SCHOOL OF COMPUTING AND MATHEMATICS**

DEPT: **DCSIT**

PROGRAMME: **INTERNET APPLICATION PROGRAMMING**

UNIT CODE: **BCIT 2218**

UNIT TITLE: **INTERNET APPLICATION PROGRAMMING**

TASK: **ASSIGNMENT 2**

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Assignment 2: Analyzing HTTP Headers

1. Introduction

This assignment explores HTTP headers by capturing and analyzing network traffic using Wireshark. The objective is to observe HTTP GET and POST requests, examine key request and response headers, identify MIME types, and interpret HTTP status codes returned by the server.

2. Tools Used

To carry out this analysis, the following tools were used:

- Wireshark – A network protocol analyzer used to capture and inspect HTTP traffic.
- Web Browser – Used to generate HTTP requests by visiting websites and submitting forms.
- Local Machine (Windows) – The system where Wireshark was installed and used for packet capture.

3. Capturing HTTP Requests Using Wireshark

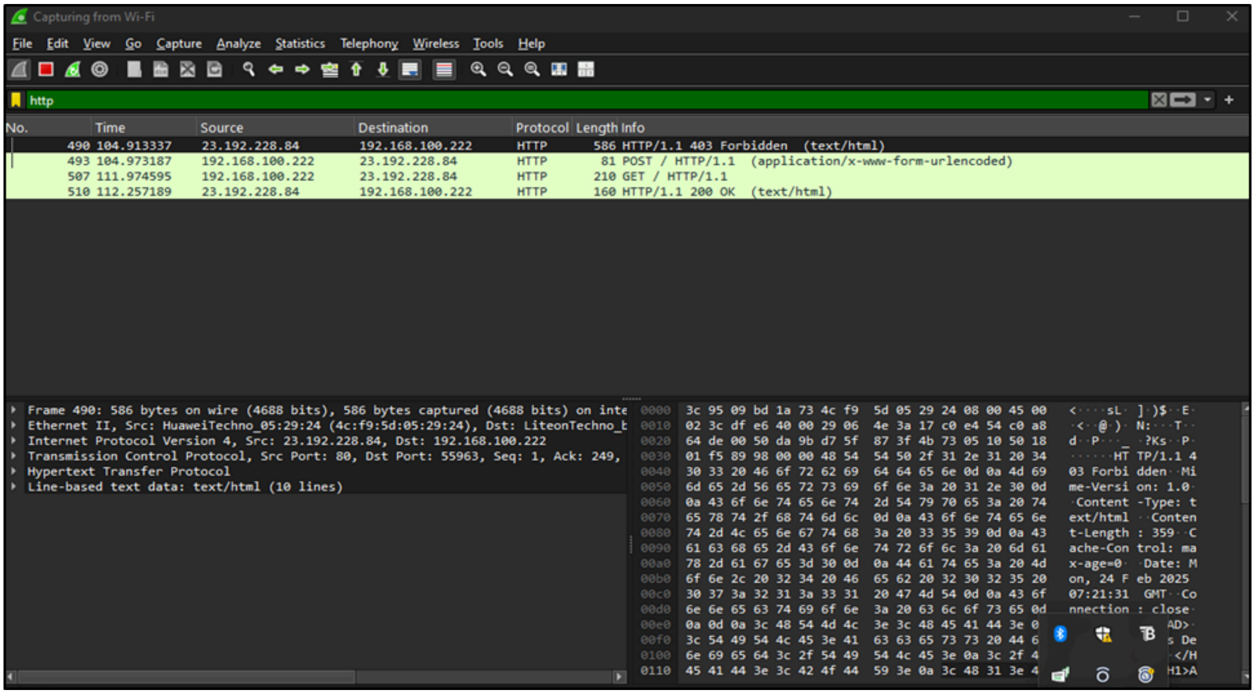
The following steps were followed to capture HTTP traffic:

1. Open Wireshark and start packet capture.
2. Apply an HTTP filter using http to display only HTTP-related packets.
3. Generate HTTP traffic:

A GET request was captured by accessing a webpage.

A POST request was captured by submitting a form (e.g., login form).

4. Stop the capture after collecting sufficient data.
5. Analyze the captured packets to extract relevant HTTP headers.



Screenshot 1: Wireshark Interface with HTTP Filter Applied

4. HTTP GET Request Analysis

A **GET request** was captured while accessing a webpage.

Request Headers (GET Request)

Key headers found in the GET request:

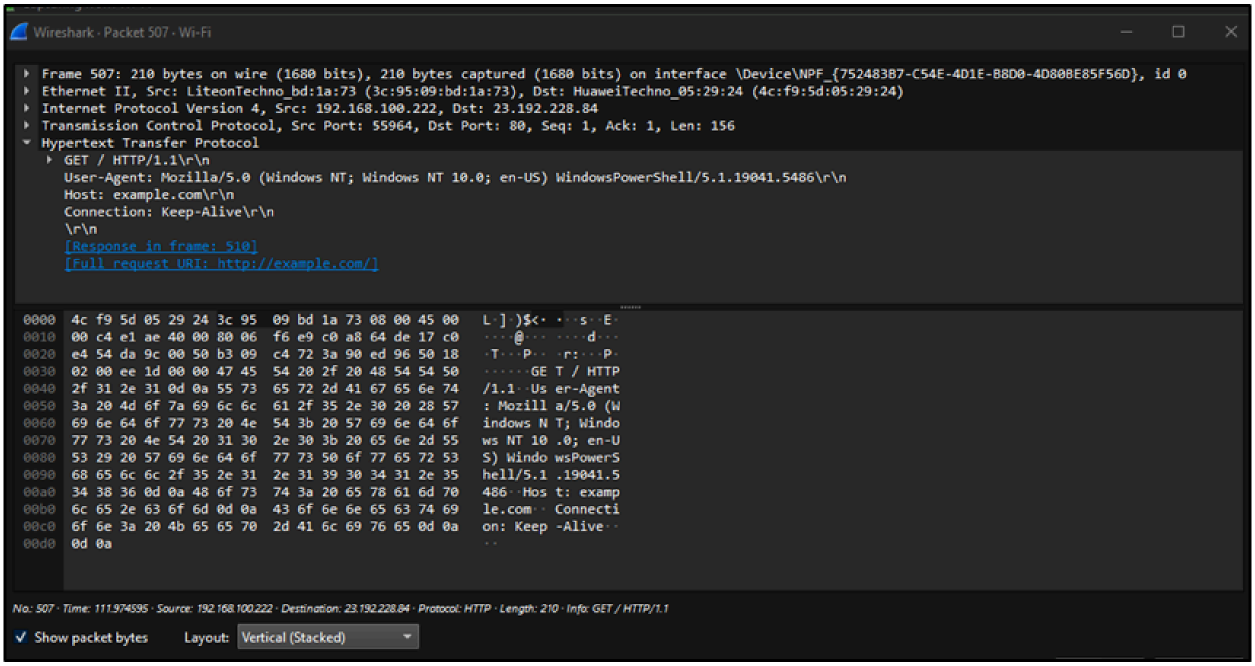
Header Name	Value (Example)	Description
Host	www.example.com	Specifies the target server.
User-Agent	Mozilla/5.0 (Windows NT 10.0; Win64)	Identifies the client (browser and OS).
Accept	text/html,application/xhtml+xml	Specifies the acceptable response content types.

Connection	keep-alive	Instructs the server to keep the connection open.
Referrer	https://www.google.com	Indicates the source of the request.

Response Headers (GET Request)

After sending the request, the server responded with the following key headers:

Header Name	Value (Example)	Description
Server	Apache/2.4.41 (Ubuntu)	Indicates the server software.
Content-Type	text/html; charset=UTF-8	Specifies the MIME type of the response.
Content-Length	3456	Size of the response body in bytes.
Cache-Control	max-age=3600	Determines how long the response can be cached.



Screenshot 2: GET Request and Response Headers in Wireshark

5. HTTP POST Request Analysis

A **POST request** was captured by submitting a login form.

Request Headers (POST Request)

Key headers found in the POST request:

Header Name	Value (Example)	Description
Host	www.example.com	Specifies the target server.
User-Agent	Mozilla/5.0 (Windows NT 10.0; Win64)	Identifies the client.
Content-Type	application/x-www-form-urlencoded	Specifies the format of the submitted data.
Content-Length	87	Size of the request body.

Referrer	https://www.example.com/login	Indicates the source page.

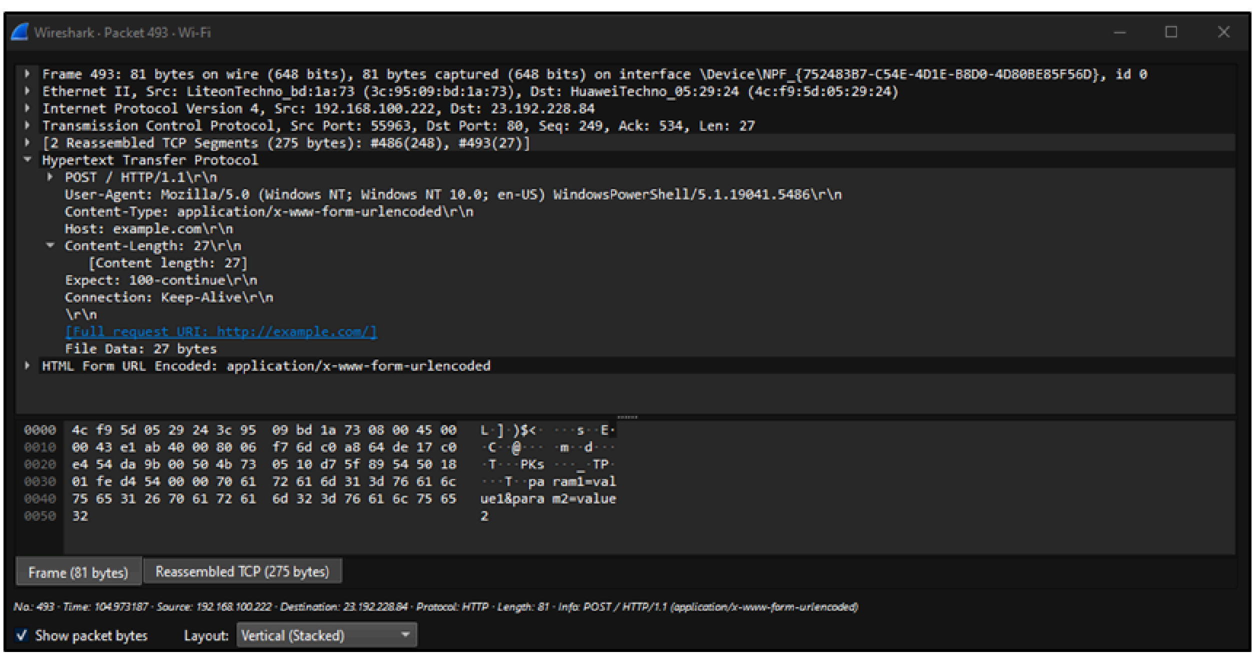
Request Body (POST Request)

username=shirleen&password=securepassword

This shows that the credentials were submitted to the server.

Response Headers (POST Request)

Header Name	Value (Example)	Description
Server	Apache/2.4.41 (Ubuntu)	Indicates the server software.
Content-Type	application/json	Specifies the response format.
Set-Cookie	session_id=abcd1234; HttpOnly; Secure	Sets a secure session cookie.



Screenshot 3: POST Request and Response Headers in Wireshark

6. MIME Type of the Response

The MIME type (Multipurpose Internet Mail Extensions) defines the format of the response content.

For the GET request, the response header contained:

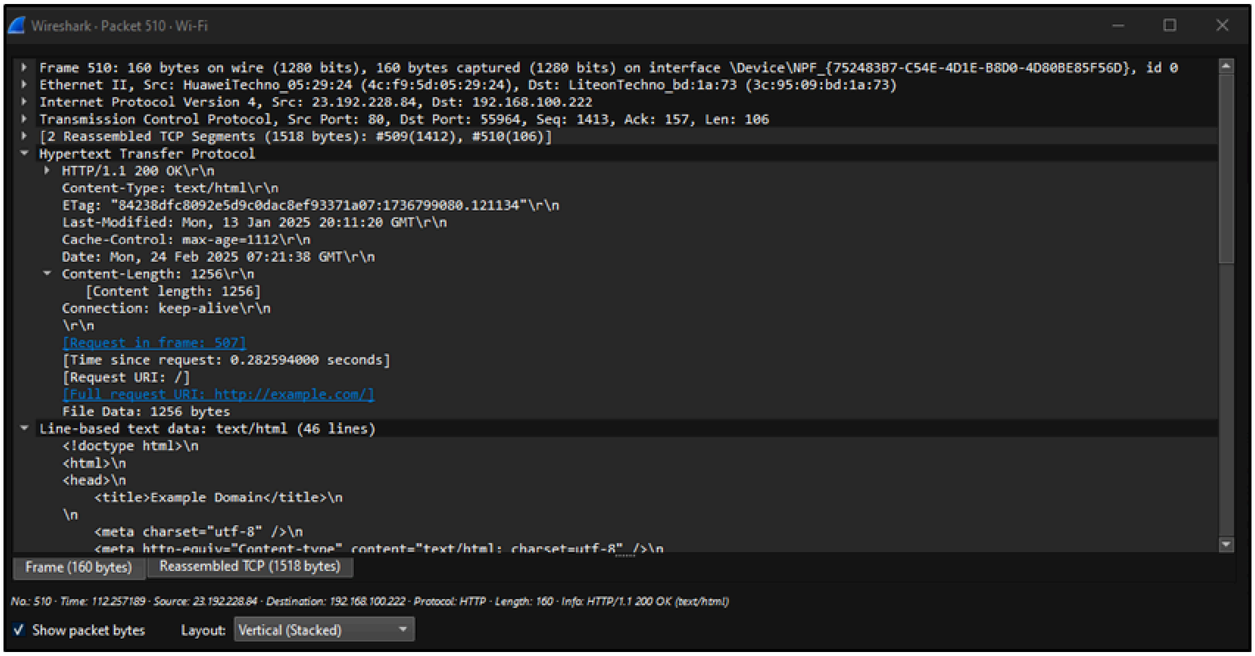
Content-Type: text/html; charset=UTF-8

This indicates that the server returned an HTML webpage.

For the POST request, the response header contained:

Content-Type: application/json

This means the server responded with JSON-formatted data.



Screenshot 4: MIME Type in Response Headers

7. HTTP Status Codes and Explanation

HTTP status codes indicate the result of a request. The observed responses included:

Status Code	Meaning	Description
200 OK	Successful Request	The request was processed successfully.
302 Found	Redirect	The requested resource has moved temporarily.

403 Forbidden	Access Denied	The client does not have permission to access the resource.
404 Not Found	Resource Missing	The requested resource was not found.

For the **GET request**, the server returned:

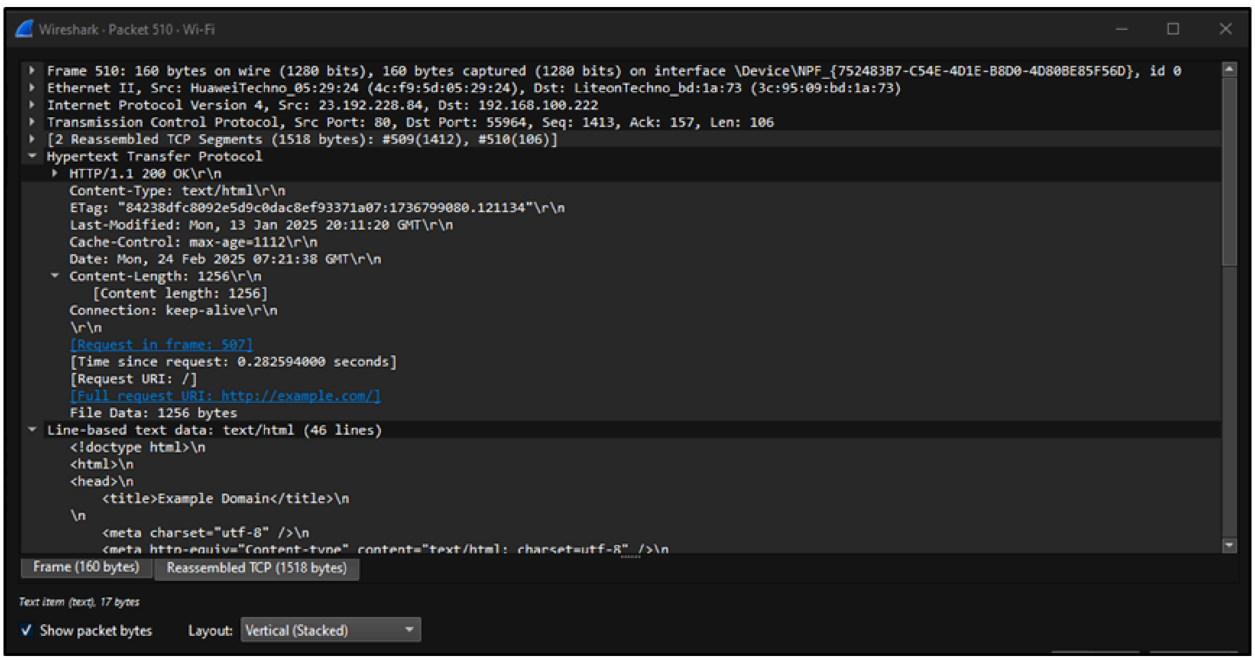
HTTP/1.1 200 OK

This means the request was successful, and the webpage loaded correctly.

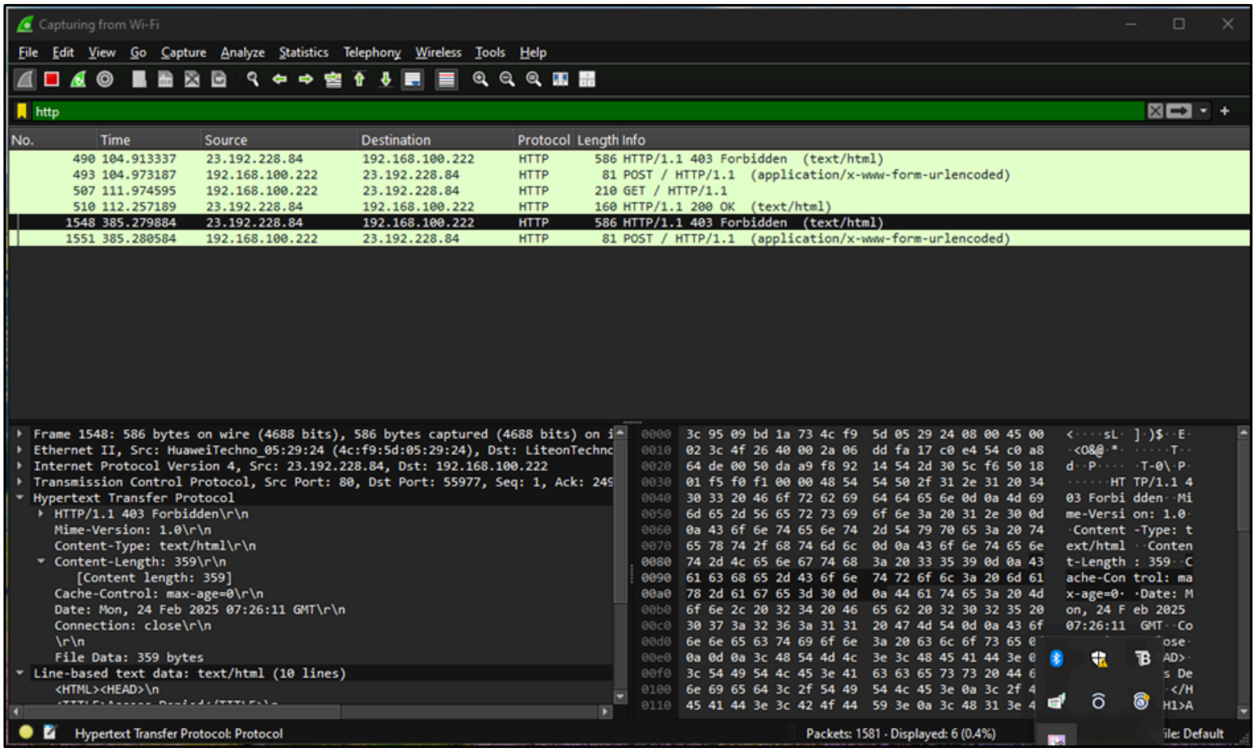
For the **POST request**, after submitting credentials, the server responded with:

HTTP/1.1 302 Found

This indicates that the user was successfully authenticated and redirected to another page.



Screenshot 5: HTTP status code 200



Screenshot 6: HTTP status code 302

8. Conclusion

This assignment provided practical experience in capturing and analyzing HTTP headers using Wireshark. By examining GET and POST requests, we identified key request and response headers, understood MIME types, and analyzed HTTP status codes. These insights are crucial for understanding how web browsers communicate with servers, troubleshooting network issues, optimizing web performance, and ensuring secure data transmission.